CS23532 - COMPUTER NETWORKS LABORATORY

EXPERIMENT 9

AIM:

To implement SUBNETTING in CISCO PACKET TRACER simulator.

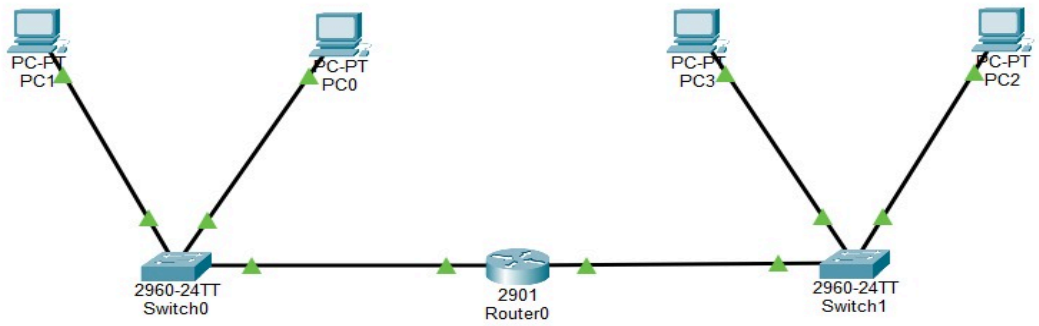
DESCRIPTION:

- **Classless Subnetting:** Classless IP subnetting allows efficient IP address utilization by using variable-length subnet masks (VLSM) rather than default classful masks.
- **Network Segmentation:** Dividing a larger network space into smaller subnets reduces broadcast domain size, enhances network performance, and improves security management.
- **Topology Design:** In this experiment, a network address space of 192.168.1.0/24 is divided into /27 subnets (providing 8 subnets with 30 usable host IPs each) connected across routers and switches.

STEP-BY-STEP PROCEDURE

Task 1: Creating Network Topology in Packet Tracer

1. Open Cisco Packet Tracer and create a new workspace.
2. Add network devices: **1 Router** (Cisco 2901), **2 Switches** (Cisco 2960), and **4 Generic PCs**.
3. Connect devices using **Copper Straight-Through Cables**:
    - PC0 (FastEthernet0) → Switch 0 (FastEthernet0/1)
    - PC1 (FastEthernet0) → Switch 0 (FastEthernet0/2)
    - Switch 0 (FastEthernet0/24) → Router0 (GigabitEthernet0/0)
    - PC2 (FastEthernet0) → Switch 1 (FastEthernet0/1)
    - PC3 (FastEthernet0) → Switch 1 (FastEthernet0/2)
    - Switch 1 (FastEthernet0/24) → Router0 (GigabitEthernet0/1)



Task 2: Router Interface Configuration

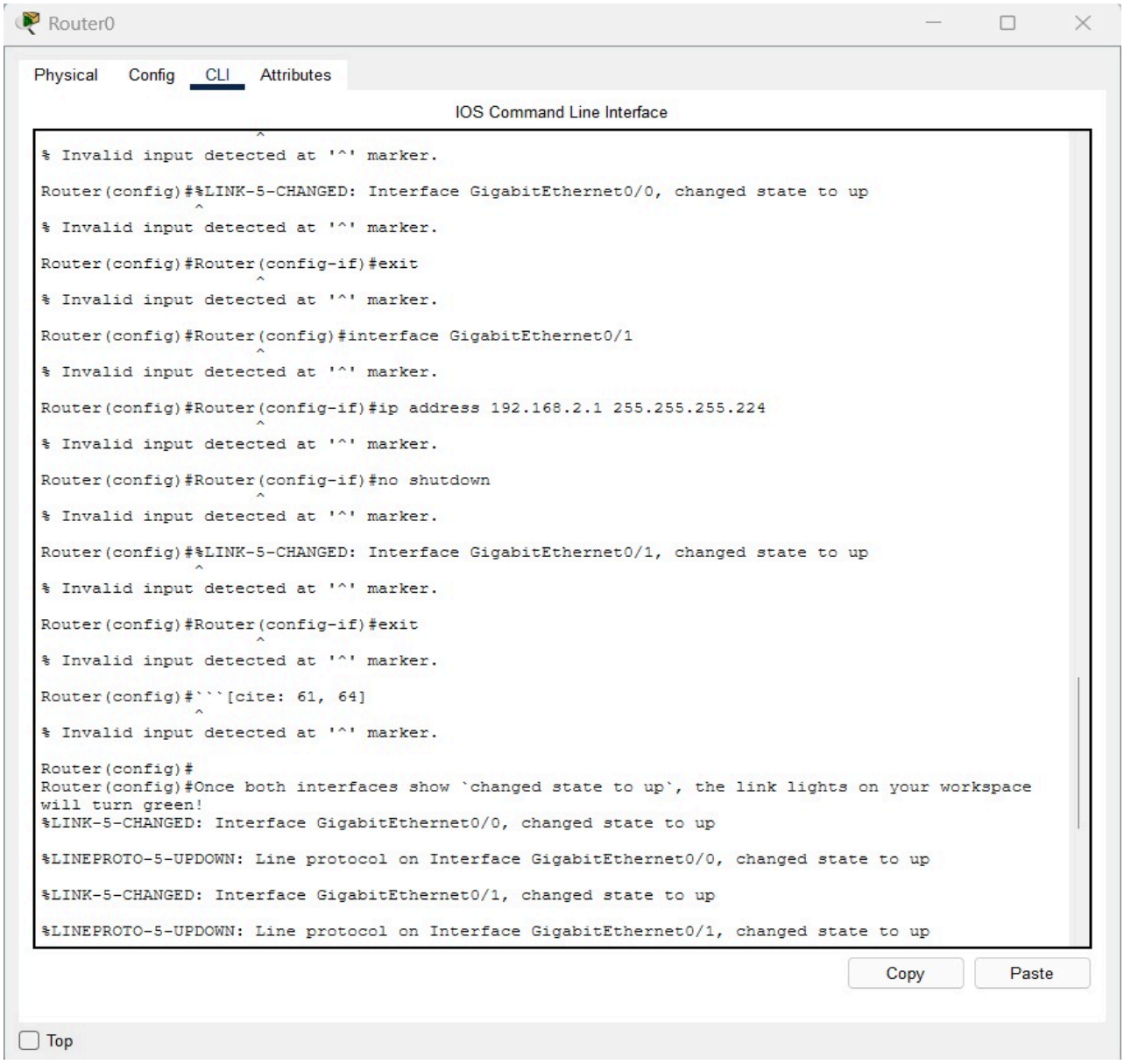
1. Open Router0, go to the **CLI** tab, and enter configuration mode:
2. Plaintext

```
Shell
enable
configure terminal

interface GigabitEthernet0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit

interface GigabitEthernet0/1
ip address 192.168.2.1 255.255.255.224
no shutdown
exit
```

Observe interface status changing to UP (%LINK-5-CHANGED: Interface... state to up).

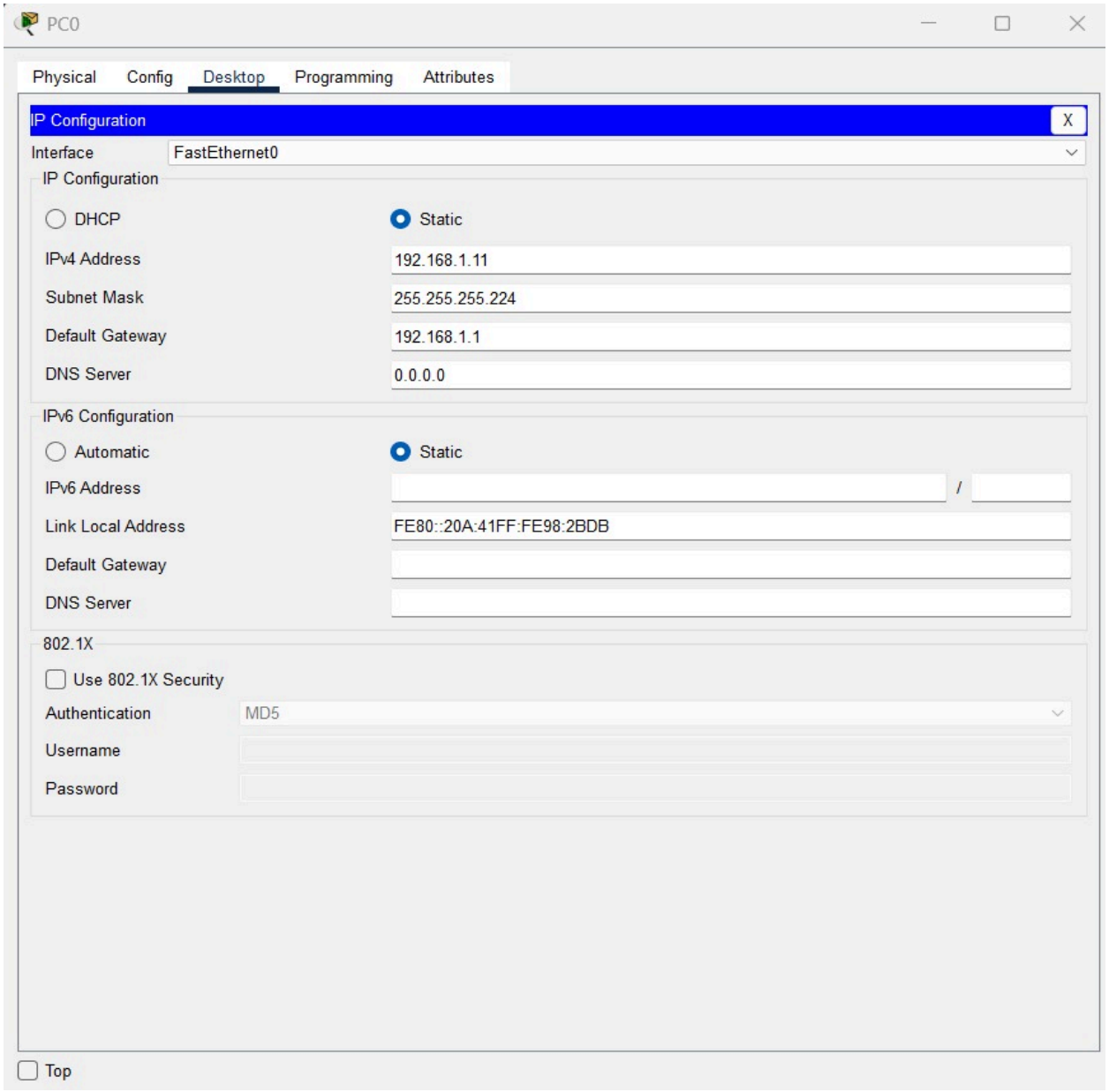


Task 3: Host IP Configuration

Click each PC, open **Desktop** → **IP Configuration**, select **Static**, and assign /27 network parameters:

- **Subnet 1 (192.168.1.0/27):**
    - **PC0:** IP: 192.168.1.11 | Mask: 255.255.255.224 | Gateway: 192.168.1.1
    - **PC1:** IP: 192.168.1.12 | Mask: 255.255.255.224 | Gateway: 192.168.1.1
- **Subnet 2 (192.168.2.0/27):**

- **PC2:** IP: 192.168.2.11 | Mask: 255.255.255.224 | Gateway: 192.168.2.1
- **PC3:** IP: 192.168.2.12 | Mask: 255.255.255.224 | Gateway: 192.168.2.1



Task 4: Connectivity Testing & Verification

1. Open **Command Prompt** on PC0 (192.168.1.11).
2. Verify intra-subnet connectivity:

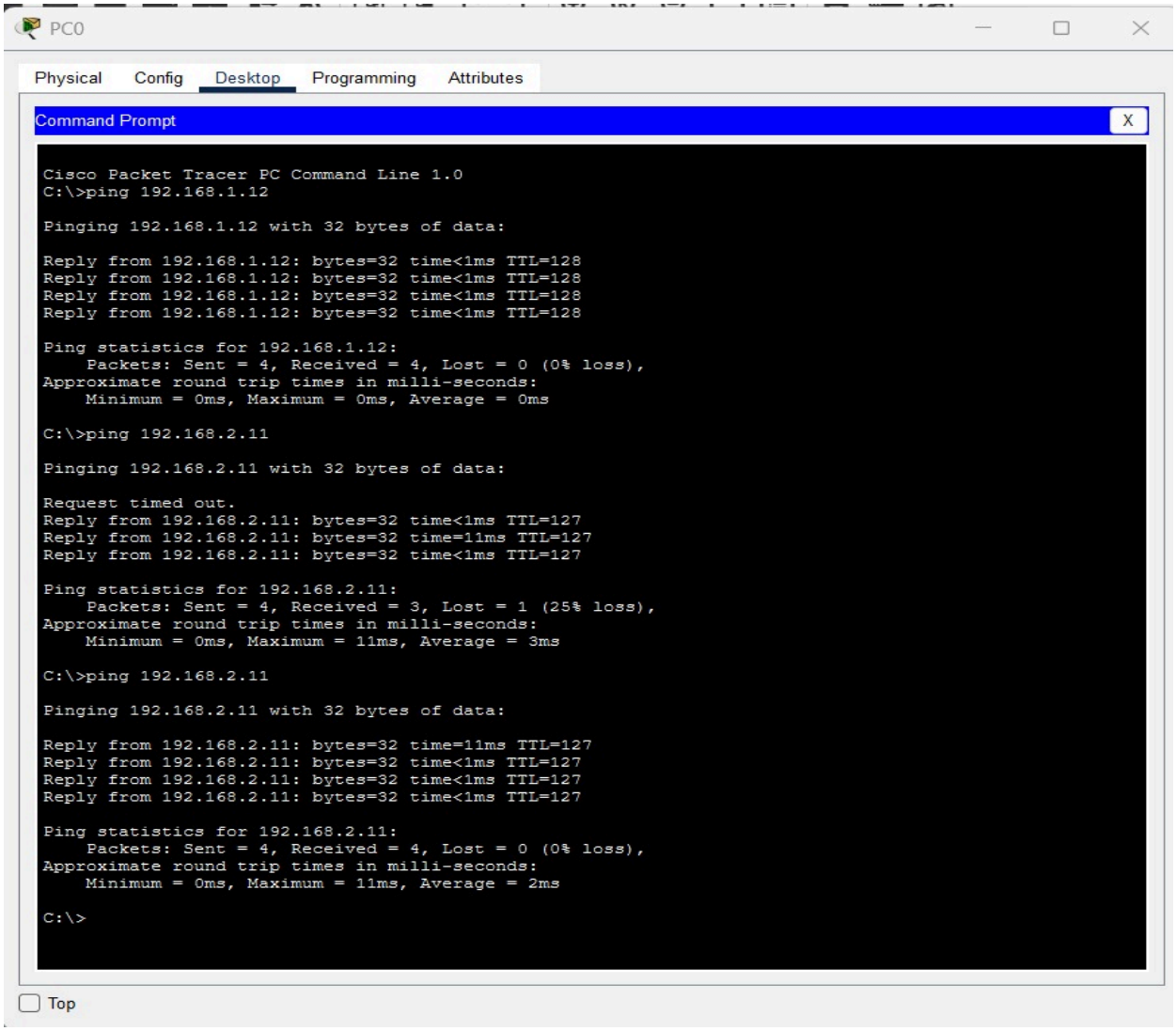
None

```
ping 192.168.1.12
```[cite: 63]

3. Verify inter-subnet connectivity via router gateway:

None

```
ping 192.168.2.11
```[cite: 63]



SUBNETTING ADDRESSING TABLE

Device	Interface	Assigned IP Address	Subnet Mask (VLSM /27)	Default Gateway	Subnet Range
Router0	Gig0/0	192.168.1.1	255.255.255.224	*N/A (Gateway)*	192.168.1.0/27
Router0	Gig0/1	192.168.2.1	255.255.255.224	*N/A (Gateway)*	192.168.2.0/27
PC0	Fa0	192.168.1.11	255.255.255.224	192.168.1.1	192.168.1.0/27
PC1	Fa0	192.168.1.12	255.255.255.224	192.168.1.1	192.168.1.0/27
PC2	Fa0	192.168.2.11	255.255.255.224	192.168.2.1	192.168.2.0/27
PC3	Fa0	192.168.2.12	255.255.255.224	192.168.2.1	192.168.2.0/27 [cite: 61]

Result: The implementation of Subnetting in Cisco Packet Tracer simulator has been constructed, configured, and tested successfully.